US DEFENCE INVESTMENT

The Misallocated Trillion

Why Defence Capital Flows to the Wrong Layer

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IDEA IN BRIEF

THE PUZZLE The US defence budget exceeds $880 billion, and private capital is flooding into defence technology. Yet capital allocation follows platform logic—which aircraft, which drone, which satellite—rather than architecture logic. The result is systematic overinvestment in technically impressive systems and systematic underinvestment in the integration, sustainment, and coordination capacity that determines whether those systems produce strategic effect.

THE EXPLANATION Defence capital markets are mispricing defense architecture. The binding constraint on US military effectiveness is not technology—it is the institutional and industrial infrastructure that converts technology into fielded, sustained, adaptable capability. Capital that flows into the platform layer while the bottleneck sits in the integration layer is chasing returns in a non-binding constraint.

THE FRAMEWORK This uses a binding constraint analysis, applied to capital allocation. Investment returns are highest when capital addresses the weakest layer in a system. In the US defence-industrial base, that layer is not R&D or platform design—it is the mid-tier integration capacity, sustainment infrastructure, procurement speed, and allied industrial depth that convert R&D into strategic outcomes.

THE IMPLICATION A $200 billion reallocation guided by architecture logic would look radically different from current defence investment patterns, and the investors who understand this structural shift will capture it. Similarly, those who don't will fund capability that arrives too late and costs too much.

The defence investment thesis of the moment is seductive in its simplicity. Autonomy; artificial intelligence; hypersonics; space. Hundreds of billions of dollars in public and private capital are chasing the next transformative platform with the logic to identify the technology that will define the next conflict, fund the platform that delivers it, and back the firm that wins the contract.

This logic is not wrong, but it is incomplete in a way that systematically misallocates capital.

The 2025 Kashmir aerial exchange, Turkey's drone campaigns in Libya, Nagorno-Karabakh, and Ukraine, South Korea's rapid emergence as a global defence exporter—these episodes share a common lesson that the investment world has not yet absorbed:

The states generating the highest return on defence investment are not the ones spending the most or fielding the most advanced technology; they are the ones with the most coherent industrial architecture for converting investment into sustained, adaptable capability.

Capital markets have not absorbed this lesson, meaning that the dominant analytical framework remains platform-centric: which aircraft, which missile system, which autonomy stack.

What is missing is an architecture-level view—one that identifies where in the defence-industrial system the binding constraints actually sit, and therefore where capital generates the highest marginal return.

THE PLATFORM TRAP

Defence capital allocation today—both public procurement and private investment—operates on platform logic. The process is as follows: identify the next critical capability domain (stealth, directed energy, autonomous systems), fund the platform that delivers it, and bet on the prime contractor or the startup that wins the programme of record.

This approach produces extraordinary technology while producing platforms such as the F-35: a program whose lifetime cost now exceeds $1.7 trillion, whose development-to-full-rate-production timeline stretched beyond two decades, and whose unit economics remain

an order of magnitude above what coordinated mid-tier producers achieve for broadly comparable mission sets.

The platform trap is not that these investments are bad; consider that the F-35 is a remarkable aircraft. Instead, the trap is that platform-level investment absorbs capital and institutional attention at the expense of the layer that actually determines aggregate strategic output. Every dollar and every year of programme management attention directed at the next exquisite system is a dollar and a year not invested in the infrastructure that determines whether any system—exquisite or otherwise—actually converts into deployable, sustained, adaptable military power.

The result is a defence-industrial base that is simultaneously the most technically advanced in the world and one of the least efficient at converting investment into fielded capability per unit time.

THE BINDING CONSTRAINT IS NOT TECHNOLOGY

The binding constraint framework, developed formally in the companion paper on coordination architectures, holds that system performance is determined by the weakest layer—not the average quality across layers. Improving non-binding layers produces negligible gains in total system output. Only investment that addresses the binding constraint moves the system forward.

Applied to the US defence-industrial base, this framework reveals a striking misalignment between where capital flows and where the constraints actually sit.

The US Defence-Industrial Stack

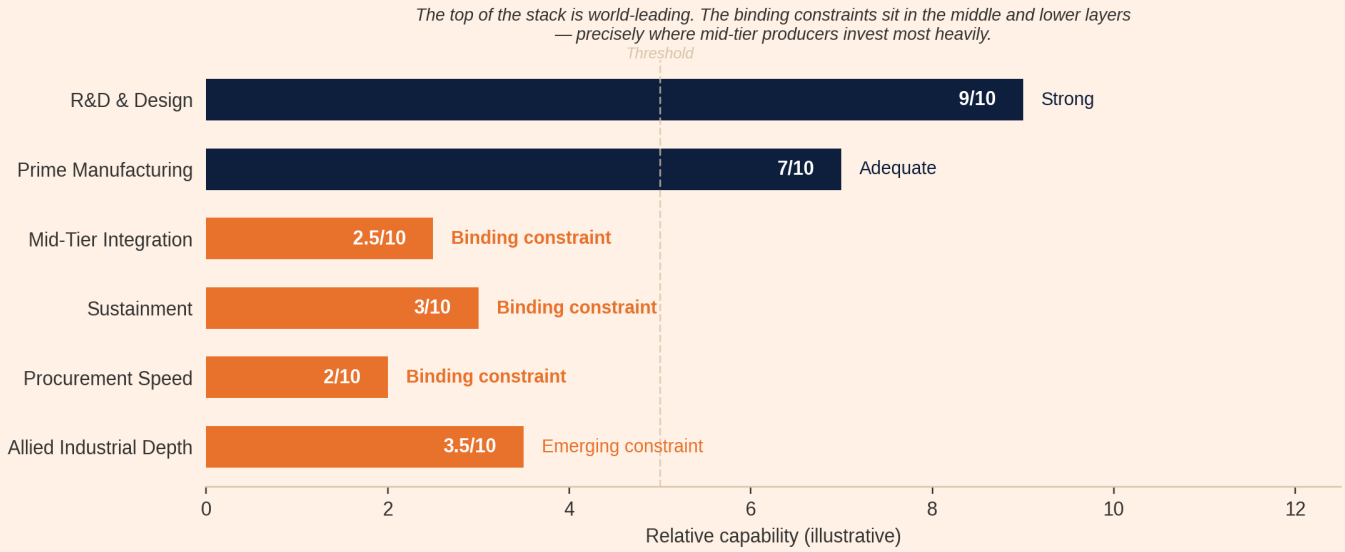

Figure 1. *The US defence-industrial stack scored by layer. The top of the stack is world-leading. The binding constraints sit in the middle and lower layers—precisely where mid-tier producers invest most heavily. Author's assessment; scoring is illustrative but directionally robust.*

The top two layers of the stack—R&D and prime manufacturing—are world-leading. American research institutions, national laboratories, and the DARPA ecosystem remain unmatched. The five remaining prime contractors possess extraordinary production capacity. These are *not* the problem.

The problem sits in the middle and lower layers: the mid-tier integration capacity that once allowed competitive prototyping, rapid systems integration, and agile modification has been hollowed out by three decades of consolidation. In 1993, the US defence-industrial base contained fifty-one firms capable of acting as prime contractors or lead integrators; today there are five. The firms that once sat between the primes and the subcontractor base—performing the systems integration work that Pakistan's PAC Kamra and Turkey's Baykar now perform—have been absorbed or eliminated.

Sustainment, which accounts for roughly seventy percent of a weapon system's lifecycle cost, is treated as an afterthought in both Pentagon budgeting and private investment allocation. This means that the states fielding the highest-performing forces per dollar spent—including Pakistan and South Korea—have built sovereign sustainment capacity as a first-order investment priority. The US has not.

Procurement architecture—the regulatory and institutional machinery through which requirements become contracts and contracts become fielded systems—has calcified into a structure designed for Cold War timelines. The DFARS and FAR regulatory frameworks, the JCIDS requirements process, and the milestone-driven acquisition lifecycle impose institutional friction that no amount of platform-level investment can overcome.

Allied industrial depth—the capacity to co-produce, co-sustain, and co-integrate with partner nations—remains structurally underbuilt. American arms sales are designed as finished-product transactions that create dependency, not as shared industrial ventures that create distributed resilience.

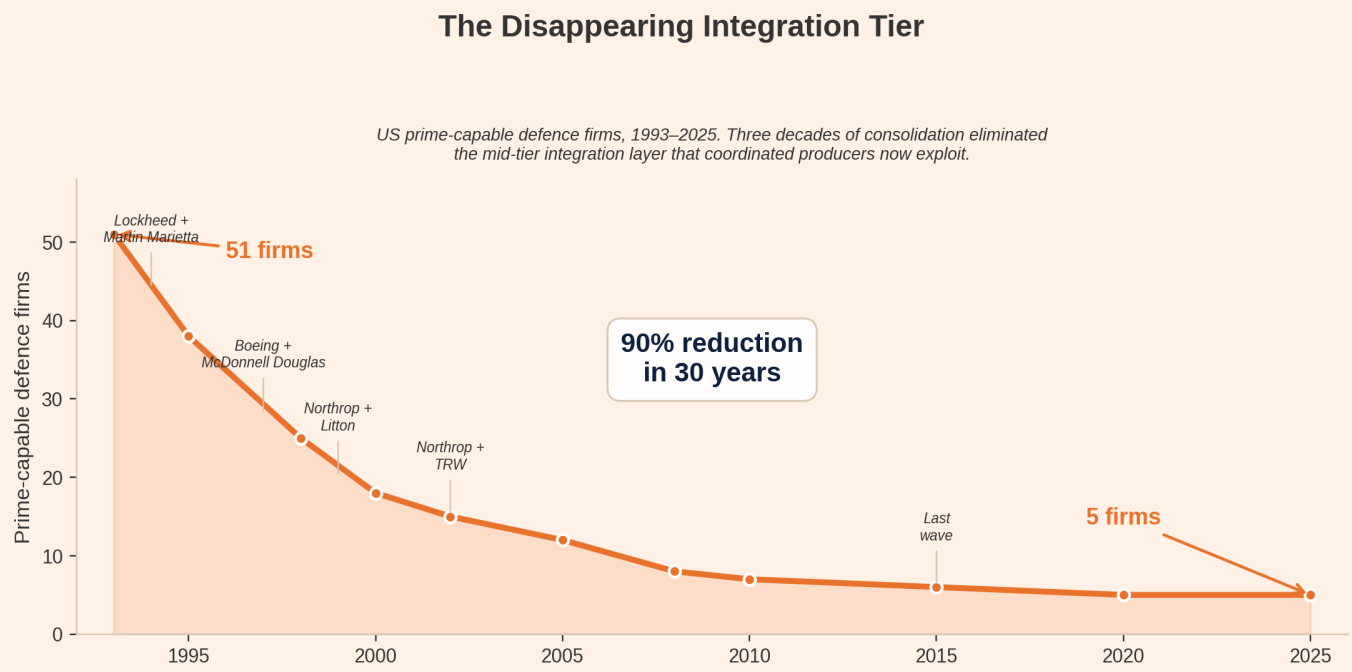

Figure 2. *US prime-capable defence firms, 1993–2025. Three decades of consolidation eliminated the mid-tier integration layer that coordinated producers now exploit. Sources: DoD historical records; GAO consolidation reports; author estimates.*

THE CONVERSION RATE

The metric that exposes this misalignment most clearly is not total spending but conversion rate: the ratio of investment to fielded, sustained, adaptable capability per unit time.

On this metric, the US performs remarkably poorly relative to coordinated mid-tier producers. The F-35 required over two decades from concept to initial operational capability. In that same period, Pakistan fielded three successive block upgrades of the JF-17, each incorporating new avionics, weapons integration, and electronic warfare capabilities. Turkey developed, tested, deployed, and exported the TB2 drone system. South Korea progressed the KF-21 from design to prototype to planned initial production faster than the US has resolved the F-35's software sustainment challenges.

These are not comparable platforms in absolute technical terms, and that is precisely the point. The mid-tier producers are *not* trying to match American technical ceilings; they are optimizing for a different variable: the speed and efficiency with which investment converts into operational capability. And on that variable, they are winning.

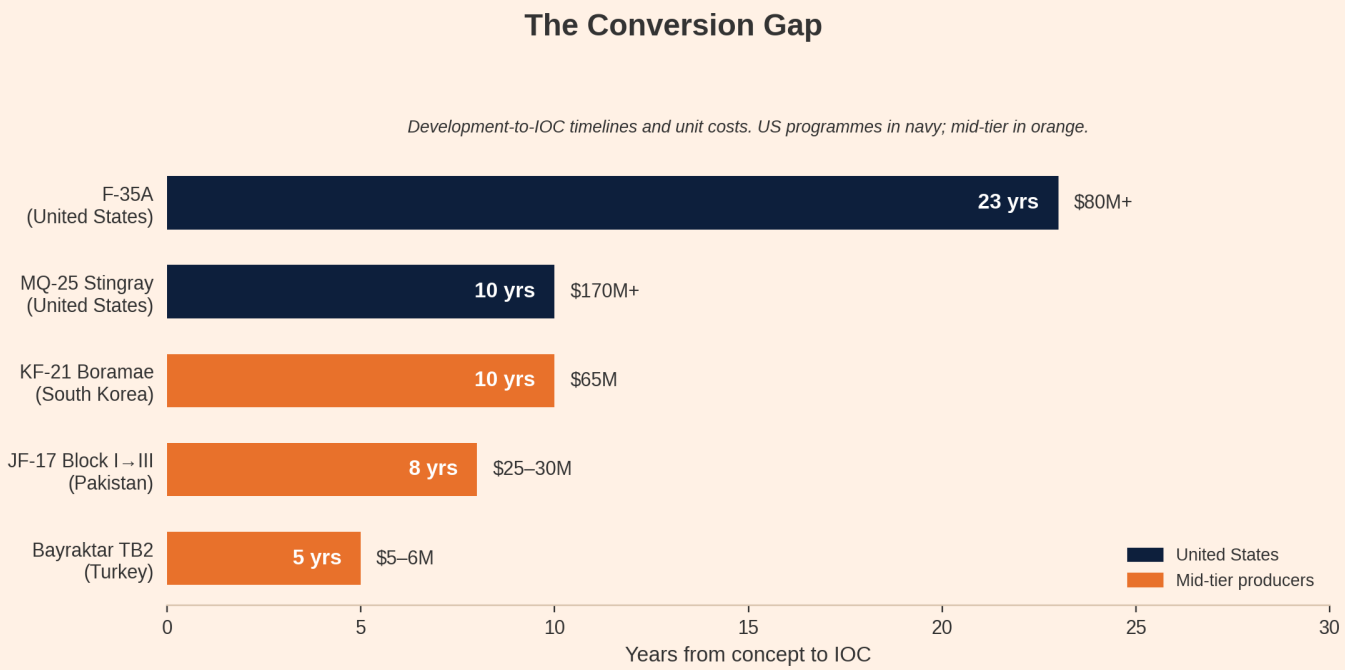

Figure 3. *Development-to-IOC timelines and unit costs across US and mid-tier programmes. The gap is not ambition—it is industrial friction. Sources: DoD Selected Acquisition Reports; manufacturer disclosures; SIPRI; author estimates.*

The mispricing is not in which platform to back, but in which layer of the system to fund. Capital that flows to the platform layer while the binding constraint sits in the integration layer is not investing—it is decorating.

WHERE THE MISPRICING SITS

The binding constraint analysis translates directly into investment language. Five areas emerge where architecture logic says capital is systematically misallocated.

Integration tier. The disappearance of mid-tier defence firms is the single largest structural weakness in the US defence-industrial base. These were the firms that performed competitive prototyping, systems integration at speed, and rapid platform modification—precisely the functions that define the coordination model now emerging among mid-tier producers. Rebuilding this layer means funding firms capable of systems integration outside the prime monopoly, open-architecture software platforms that reduce vendor lock-in, and rapid modification capacity that allows existing platforms to be upgraded on operational timelines rather than procurement timelines. DARPA's recent moves toward non-traditional contractors point in this direction but lack institutional scale.

Sustainment infrastructure. Approximately seventy percent of a weapon system's total cost is incurred after acquisition—in maintenance, spare parts, software updates, and depot-level overhaul. Yet sustainment is where budgets are cut first and where private investment is thinnest. The states achieving the highest defence return on investment have built sovereign sustainment as a first-order priority, not a residual claim on leftover budget. Capital here means predictive maintenance technology, expanded depot-level capacity, supply chain resilience programmes, and software sustainment for fielded systems that currently depend on prime contractor support at monopoly pricing.

Speed-track procurement. The US defence procurement system has one speed: slow. Creating a parallel track for what might be called "eighty-percent platforms"—systems that sacrifice the last twenty percent of technical performance for three times faster fielding—is an architectural intervention, not merely a policy

preference. This does not replace the F-35 or B-21. It creates a second lane for systems where iteration speed matters more than absolute technical edge. Capital here means funding mechanisms, contracting vehicles, and programme offices designed for three-year development cycles rather than twenty-year ones.

Allied co-production capacity. The US treats arms sales as finished-product transactions. The growing ideology in that is proving successful in the emerging mid-tier model treats them as shared industrial architecture. Converting allied relationships from dependency instruments—"buy our F-16, depend on our sustainment"—into co-production ventures creates distributed integration capacity that serves both American and allied needs. The partners best positioned for this—South Korea, Japan, Australia, Poland—already possess substantial industrial capacity. What is missing is the institutional architecture to connect it.

What you stop funding. Another generation of exquisite platforms competing for the same five primes' capacity. Not because frontier platforms are unnecessary—they are essential for peer competition—but because the marginal dollar invested in the sixth next-generation programme produces less strategic return than the first dollar invested in rebuilding the integration tier. The US needs a frontier track for great-power competition and a speed track for everything else. Running the entire portfolio through the frontier track is the architectural failure.

The $200 Billion Reallocation

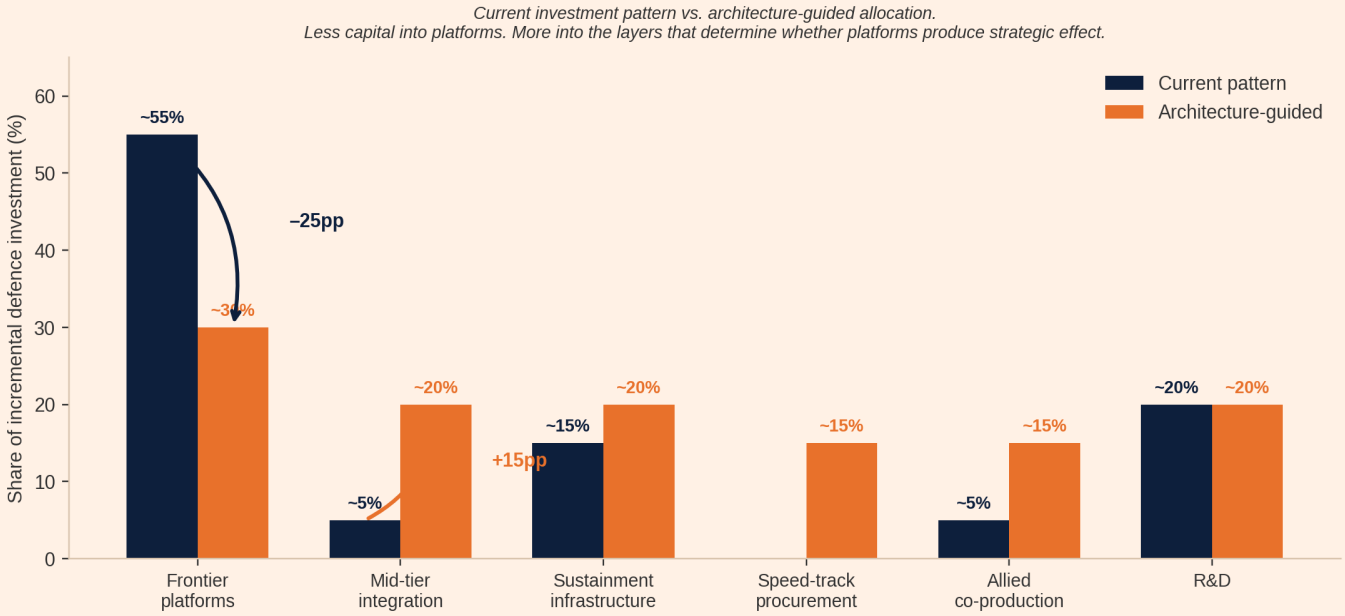

Figure 4. *Current investment pattern vs. architecture-guided allocation. Less capital into platforms. More into the layers that determine whether platforms produce strategic effect. Author's framework; shares are illustrative and directional, not prescriptive.*

The $200 Billion Portfolio Reallocation

Current investment portfolio vs. architecture-guided allocation. Illustrative shares of incremental defence investment.

INVESTMENT AREA	CURRENT PATTERN	ARCHITECTURE-GUIDED	RATIONALE
Frontier platforms	~55%	**~30%**	Still essential for peer competition. But marginal returns are declining.
Mid-tier integration	~5%	**~20%**	The hollowed-out layer. Rebuild firms capable of systems integration outside the prime monopoly.
Sustainment infrastructure	~15%	**~20%**	70% of lifecycle cost. Predictive maintenance, depot capacity, software sustainment.
Speed-track procurement	~0%	**~15%**	80% platforms on 3-year cycles. A parallel track, not a replacement.
Allied co-production	~5%	**~15%**	Shared integration capacity with South Korea, Japan, Australia, Poland.
R&D	~20%	**Unchanged**	Already strong. Not the binding constraint.

Figure 5. *Author's framework. Shares are illustrative and directional, not prescriptive.*

THE MID-TIER BENCHMARK

The coordinated mid-tier producers—Pakistan, Turkey, South Korea, the UAE—are useful not as threats but as benchmarks; their capital allocation patterns reveal where the highest marginal returns sit when a state is forced by constraint to invest efficiently.

These states spend disproportionately on integration, sustainment, and iteration speed. Pakistan's entire JF-17 programme—from initial co-development through three block upgrades—has cost less than the US spends annually on F-35 sustainment alone. Turkey built Baykar's drone ecosystem for a fraction of what the MQ-25 programme will consume. South Korea's KF-21 is progressing from design to planned production in roughly the time the US takes to complete a major software update on an existing platform.

The lesson is not that the US should copy Pakistan's model. Pakistan operates under constraints—fiscal, industrial, geopolitical—that the US does not face. The lesson is that the architecture-lag framework reveals where the highest marginal returns on defence investment sit, and the mid-tier producers have found them first because constraint forced them to look in the right place.

The US, unconstrained by fiscal pressure, has been free to keep investing in the layers it does best—frontier R&D and exquisite platforms—while allowing the binding constraint layers to atrophy. This is the luxury of wealth misapplied: the ability to keep spending without confronting the question of whether spending is converting into strategic effect.

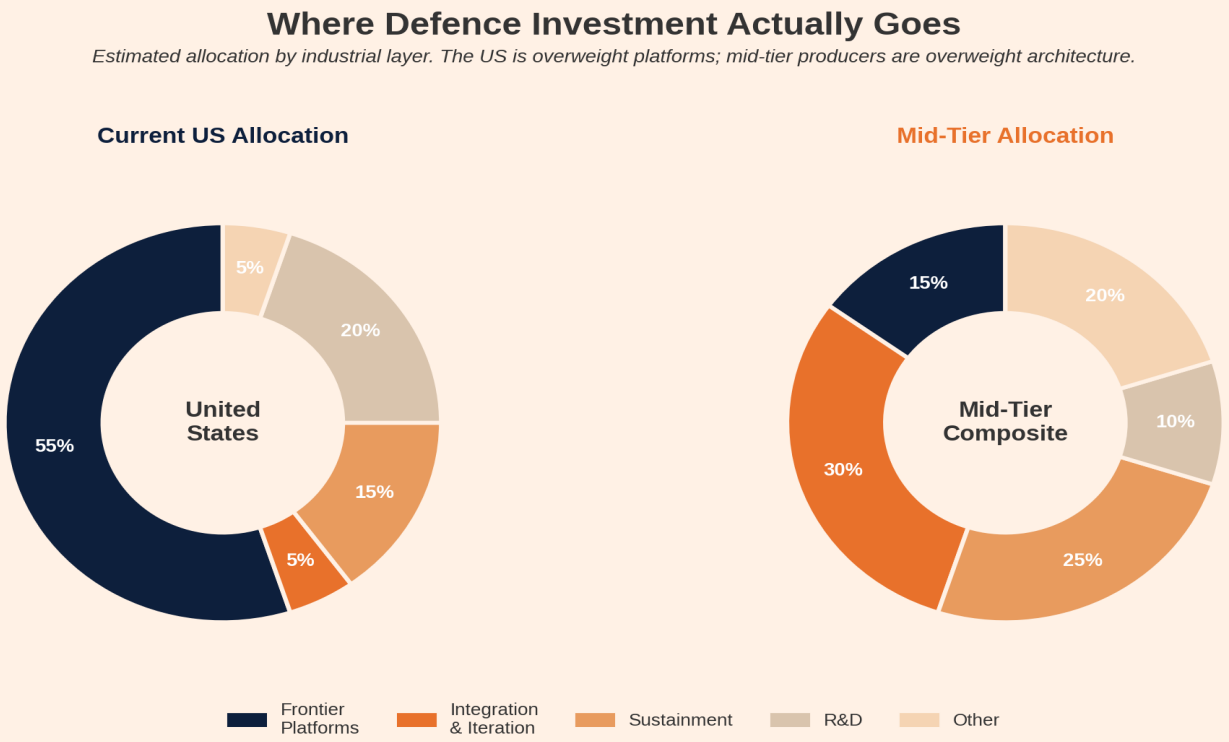

Figure 6. *Estimated defence investment allocation by industrial layer. The US is overweight platforms; mid-tier producers are overweight architecture. Sources: author estimates based on DoD budget data, SIPRI, national disclosures.*

WHAT THIS IS NOT

This is not an argument against technical superiority. In a peer competition with China, the US needs platforms that operate at or beyond the technological frontier. The F-35, the B-21, the next generation of space-based systems—these serve functions that no eighty-percent platform can replace.

Nor is this an argument that the US should reduce defence spending. The spending level is not the problem. The conversion architecture is the problem.

What the architecture-lag framework reveals is that the US needs two tracks operating simultaneously: a frontier track for peer competition, where technical ceiling matters and development timelines are acceptable, and a speed track for the rest of the threat landscape, where iteration speed, affordability, and industrial resilience matter more than absolute performance.

Running the entire portfolio through the frontier track is an institutional habit that the mid-tier rise has exposed as a structural vulnerability.

The US can afford to spend. The question is whether it can afford to spend this slowly

RETHINKING WHERE RETURNS SIT

The defence capital market is running on platform logic in an era that rewards architecture logic. The states producing the highest strategic return per dollar invested are not the ones with the best technology; rather they are the ones with the most coherent systems for converting investment into fielded, sustained, adaptable capability.

A $200 billion reallocation that follows the binding constraint would not weaken American military power, it would make American military spending do what everyone already assumes it does: produce the world's

most effective fighting force per dollar invested. Right now, on the conversion metric that actually matters, it does not.

The emerging mid-tier producers that have changed the economics of global military warfare did not discover a secret, they discovered a constraint. And because they could not spend their way around it, they were forced to build the architecture that addresses it. The US has the resources to build that architecture at a scale no mid-tier producer can match. But what it lacks is the institutional recognition that the architecture—not the platform—is where the returns sit.

If defence investors and policymakers continue to allocate capital at the platform layer while the binding constraint sits in the integration, sustainment, and procurement layers, the result will be predictable: increasingly expensive capability that arrives too late, sustains too slowly, and adapts too rigidly for a threat environment defined by speed, coordination, and industrial resilience.

The mispricing is structural, while the correction is architectural. And the window in which the US can make it—before the mid-tier producers' coordination advantage becomes self-reinforcing—is not indefinite.

This article draws on two formal frameworks: coordination architectures and adaptive bandwidth (from "Institutions as Coordination Architectures") and architecture lag and premature markets (from "Market Formation as a Systems Engineering Problem"). Contact: s@sinead.co